

Which type(s) of DNA are contained in a plasmid that can be used to transform bacteria in an experiment? DNA that is:

I. single-stranded

II. linear

III. circular

☐ A. I only [3%]

☐ B. II only [3%]

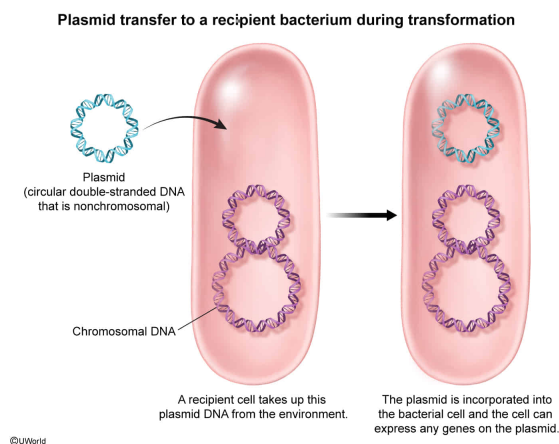
✓ ☒ C. III only [79%]

☐ D. I and II only [13%]

Correct

79% answered correctly

Explanation:



In addition to random **mutations**, bacterial genetic variation can be increased by the processes of **transduction**, **conjugation**, and **transformation**. Unlike transduction and conjugation, which require a recipient bacterium to be joined to a virus or to a donor bacterium, respectively, transformation is mediated by the recipient bacterium itself, typically in response to changes in the bacterium's environment.

During transformation, a bacterium takes up DNA directly from its extracellular environment via pores or transport proteins in the bacterium's plasma membrane. Bacteria capable of such uptake are said to be "competent," and DNA taken up by transformation can be in the form of a small fragment (eg, released from a dead bacterium nearby) or a plasmid.

The question asks for the type of DNA contained in a plasmid used to transform bacteria in an experiment. In both experimental and naturally occurring settings, **plasmid DNA is circular, double-stranded**, and not part of a chromosome (**Number III**).

(Number I) Plasmid DNA is double-stranded, *not* single-stranded.

(Number II) Plasmid DNA is circular, *not* linear.

Educational objective:

Plasmids are circular molecules of double-stranded DNA that is not part of bacterial chromosomes.

Time spent:0 sec

Subject:
Biology

QID:403676

System:
Cellular Biology